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|-------------------|----------------------------------|
| <b>Started on</b> | Saturday, 8 August 2026, 9:10 AM |
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| <b>State</b> | Finished |
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| <b>Completed on</b> | Saturday, 8 August 2026, 9:30 AM |
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|                   |                 |
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| <b>Time taken</b> | 20 mins 33 secs |
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|--------------|-----------|
| <b>Marks</b> | 3.00/3.00 |
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| <b>Grade</b> | <b>10.00</b> out of 10.00 ( <b>100%</b> ) |
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## Question 1

Correct

Mark 1.00 out of 1.00

Write a python program to find the 1-Norm of a matrix and display the results in two decimal places.

**For example:**

| Input                                 | Result |
|---------------------------------------|--------|
| <code>[[-1, 3],[3, -4],[1, 7]]</code> | 14.00  |

**Answer:** (penalty regime: 0 %)

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```
'''
Program to find 1-norm of a matrix.
Developed by: VEDHA M
RegisterNumber: 212225230292
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix = np.array(eval(input()))
result = np.linalg.norm(matrix,1)
print(result)
```

|   | Input                                       | Expected | Got  |   |
|---|---------------------------------------------|----------|------|---|
| ✓ | <code>[[-1, 3],[3, -4],[1, 7]]</code>       | 14.00    | 14.0 | ✓ |
| ✓ | <code>[[1, 2, 3],[-3,-4,-1],[9,6,1]]</code> | 13.00    | 13.0 | ✓ |

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Write a program to find L2-norm of a matrix and display the result in two decimal places.

**For example:**

| Input                     | Result |
|---------------------------|--------|
| [[1,2],[3,4]]             | 5.46   |
| [[ -1, 3],[3, -4],[1, 7]] | 8.66   |

**Answer:** (penalty regime: 0. %)

Reset answer

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```
'''
Program to find 2-norm of a matrix.
Developed by: VEDHA M
RegisterNumber: 212225230292
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix = np.array(eval(input()))
two_matrix = np.linalg.norm(matrix,2)
print("{:.2f}".format(two_matrix))
```

|   | Input                     | Expected | Got  |   |
|---|---------------------------|----------|------|---|
| ✓ | [[1,2],[3,4]]             | 5.46     | 5.46 | ✓ |
| ✓ | [[ -1, 3],[3, -4],[1, 7]] | 8.66     | 8.66 | ✓ |
| ✓ | [[2, 3],[3, 4],[1, 8]]    | 9.86     | 9.86 | ✓ |

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Write a program to find the Infinity of a matrix and display the result in two decimal places.

**For example:**

| Input                                 | Result            |
|---------------------------------------|-------------------|
| <code>[[-1, 3],[3, -4],[1, 7]]</code> | <code>8.00</code> |

**Answer:** (penalty regime: 0. %)

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```
'''
Program to find infinity of a matrix.
Developed by: VEDHA M
RegisterNumber: 212225230292
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix = np.array(eval(input()))
inf_matrix = np.linalg.norm(matrix,np.inf)
print("{:.2f}".format(inf_matrix))
```

|   | Input                                      | Expected           | Got                |   |
|---|--------------------------------------------|--------------------|--------------------|---|
| ✓ | <code>[[-1, 3],[3, -4],[1, 7]]</code>      | <code>8.00</code>  | <code>8.00</code>  | ✓ |
| ✓ | <code>[[1,2,3],[-9,-8,-3],[10,3,2]]</code> | <code>20.00</code> | <code>20.00</code> | ✓ |

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.